

Use Case LM-1.1.1.1: Initialize Engine**Scope:** Launch Mission**Primary Actor:** Flight Director**Related Use Cases:** *Use Case 1.1.1 Initialize Stage***Stake Holders & Interests:**

- Flight Director: Wants to set up the Launch via initializing all the Launch Parameters and Subsystems including:
 - Every engine
- Mission Planner: Wants the Mission Initialized with the appropriate Mission Parameters loaded

Pre-Conditions: The Mission System is running and ready to initialize and monitor the appropriate Systems and Sub-Systems**Post-Conditions:** The engines are initialized with the appropriate parameters**Flow of Events:**

Flight Director	System
1. Mission Parameters File	
	2. Finds the Engine Parameters File Name and Location
	3. Goes to the Engine Parameters File Location
	4. Opens the Engine Parameters File
	5. Reads the Engine Parameter file appropriate for the Engine (See <i>Use Case 1.1.1 Initialize Stage</i>)
	6. Initializes the Engine Data with the data from the Engine Parameter File
	7. Alerts the Flight Director the Engine is Initialized

Extensions (or Alternative Flows):

- 2a,3a If the System cannot find the location of the Engine Parameter File, then the System alerts the Flight Director of the error: indicating an illegitimate file path. The Flight Director either chooses another directory path or exits (aborts) the Mission.
- 4a If the System cannot open the Engine Parameter File, then the System alerts the Flight Director: indicating unable to open the Parameters File. The Flight Director either chooses another Parameters file, or exits (aborts) the Mission.
- 5a If the System cannot read the Engine Parameters File, then the System alerts the Flight Director: indicating unable to read the Parameters File.

The Flight Director either chooses another Parameters file or exits (aborts) the Mission.

Special Requirements:

- None

Technology & Variations List:

- There could be many types of Engine Parameters and subsequent files in different formats: the System will need to account for that.
- Regardless of the type of file, the System Data should follow a Standard.

Frequency of Occurrence:

- Everytime a Mission is planned or simulated

Open Issues:

- None